Use Case Template

CardioAI - Heart Disease Prediction System

1. Use Case Title	Hyperparameter Tuning
2. Abbreviated Title	Hyperparameter Tuning
3. Use Case Id	UC-09
4. Actors	Clinician (Primary), ML Engine (Supporting)
5. Description	The Clinician configures and runs automated hyperparameter search (grid search or Bayesian optimization) to find optimal model configurations for learning rate, dropout, hidden units, batch size, etc.
5.1. Pre Conditions	1. Training pipeline is functional (UC-07). 2. Search space is defined for target hyperparameters.
5.2. Task Sequence	1. Clinician defines the hyperparameter search space. 2. Clinician selects search strategy (Grid / Random / Bayesian). 3. ML Engine iterates through hyperparameter combinations. 4. Each combination is evaluated via cross-validation. 5. System ranks results by validation ROC-AUC. 6. System outputs the best configuration and saves the model.
5.3. Post Conditions	1. Optimal hyperparameters are identified and logged. 2. Best model corresponding to optimal config is saved.
6. Modification History	Date: September 11, 2026
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